



SAIR

Spatial AI & Robotics Lab

CSE 473/573-A

L15: TEXTURE

Chen Wang

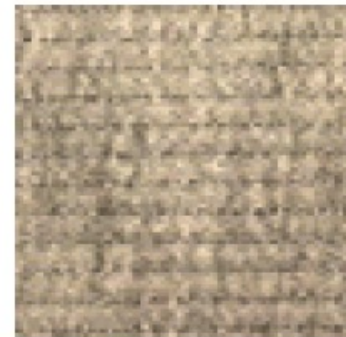
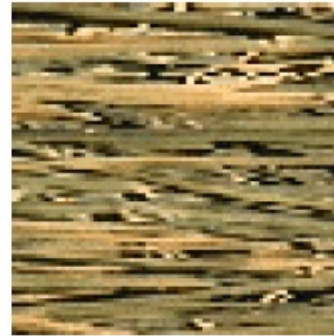
Spatial AI & Robotics Lab

Department of Computer Science and Engineering

 **University at Buffalo** The State University of New York

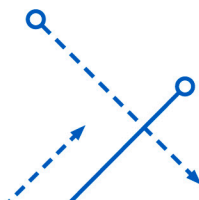
Texture

- What defines a texture?



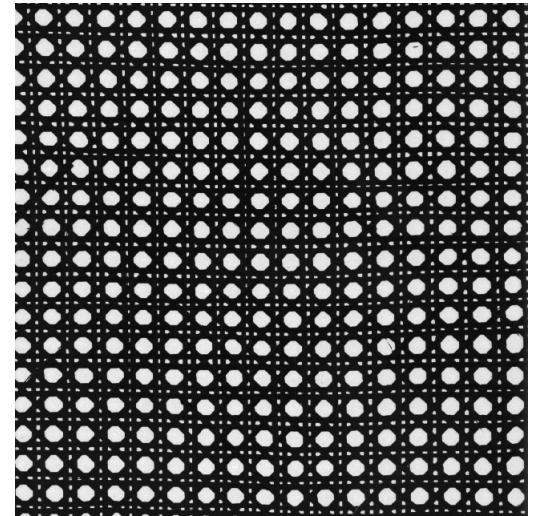
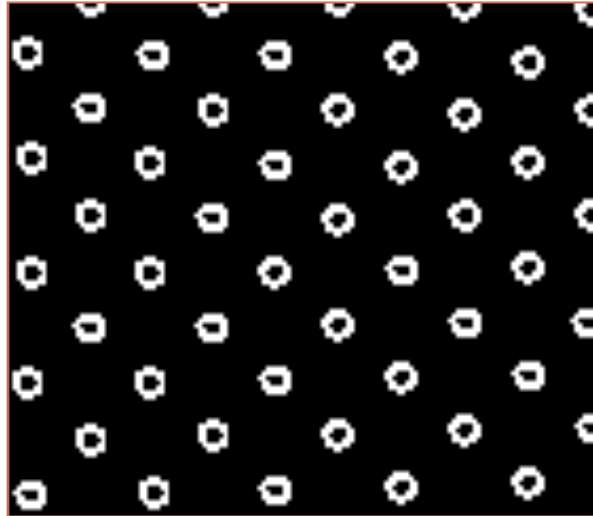
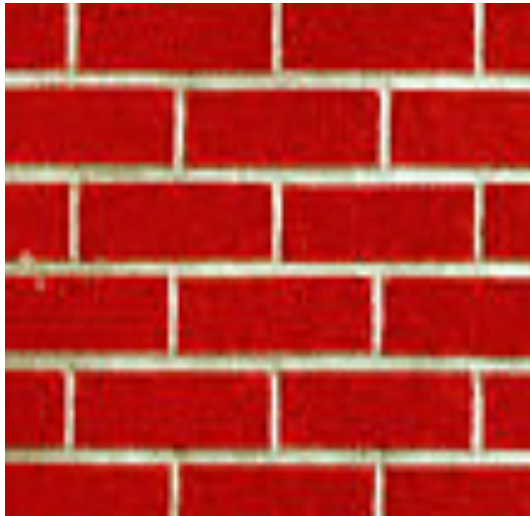
What is Texture?

- No Formal Definition
 - There will be **significant variation** in intensity levels between pixels
 - these variations perform **repetitive** patterns-homogeneous at some scale
 - **local statistics** are constant and slowly varying
- Human visual systems perceived textures as **homogeneous regions** even though they don't have the same intensity

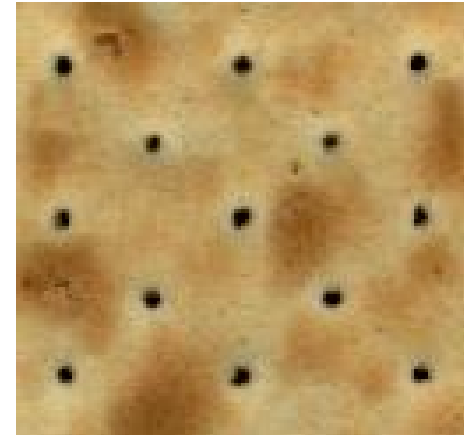
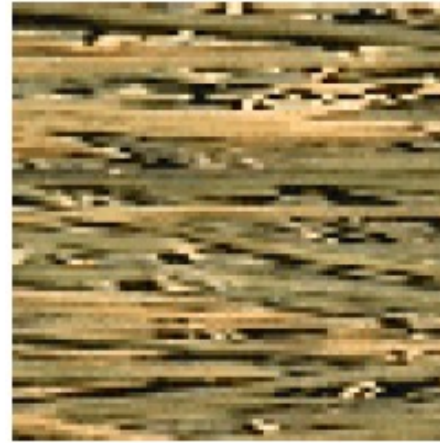


Regular (Repetitive) patterns

- We interpret this first image as a brick wall



Random patterns



Texture

- Is a property of a "group of pixels" or Area
 - a single pixel does not have texture
- It is **scale dependent**
 - at different scales textures have different properties
- It contains many possibly countless **primitive objects**
- It involves the **spatial distribution** of intensities
 - 2D histograms
 - Co-occurrence matrices

Scale

- Scale is important – consider sand
- Close up
 - “small rocks, sharp edges”
 - “rough looking surface”
 - “smoother”
- Far Away
 - “one object
 - $\Rightarrow$ brown/tan color”



How would you describe a Texture?

- Coarseness
 - Roughness
 - Direction
 - Frequency
 - Uniformity
 - Density
-
- How do you describe
 - dog fur, cat fur, wood grain, or cloth?

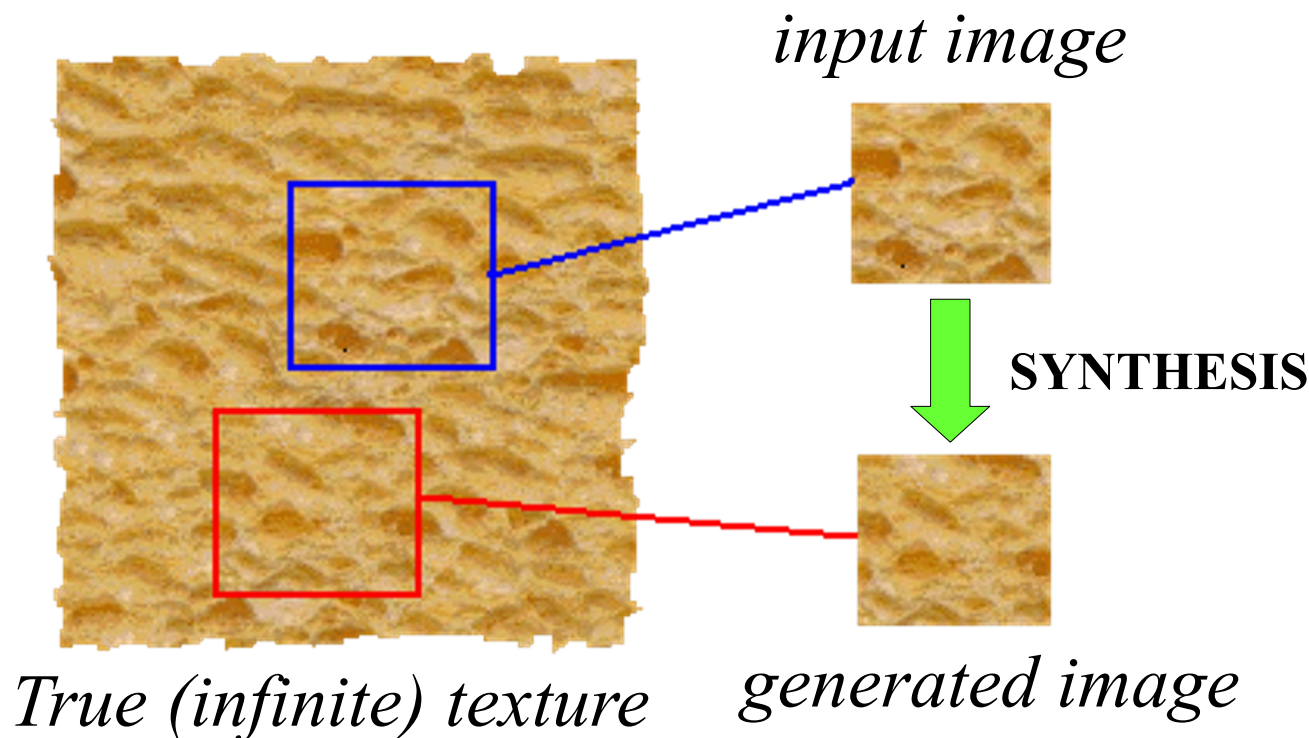
Textures has different instances.

- All images are different instances of the same texture
- We can differentiate between them, but they seem generated by the same process.



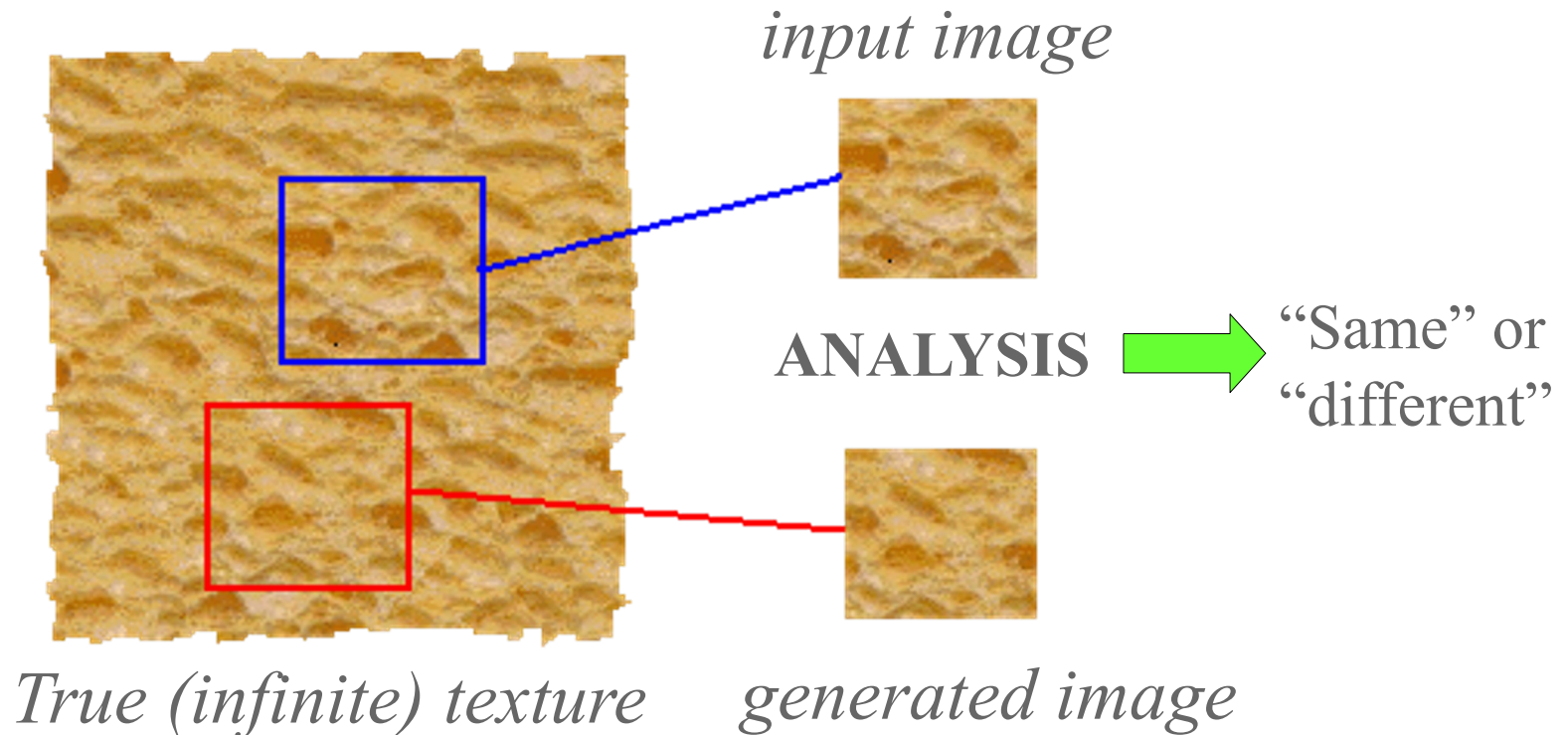
The Goal of Texture Synthesis

- Given a finite sample of some texture, the goal is to synthesize other samples from that same texture
- The sample needs to be "large enough"



The Goal of Texture Analysis

- Compare textures and decide if they're made of the same "stuff".



Scale: objects vs. texture

- The same thing can occur as texture or an object
 - depending on the **scale** we are considering.



Why analyze texture?

Importance to perception:

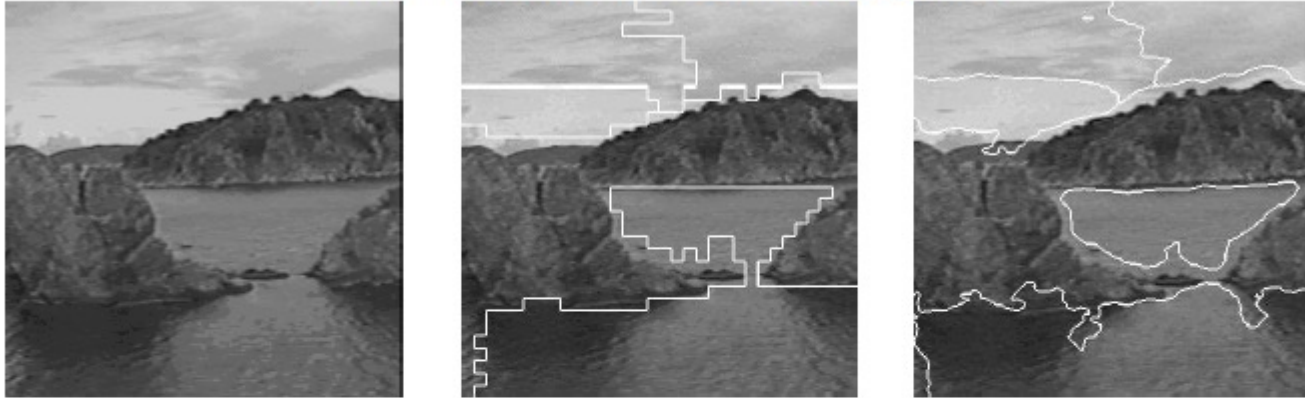
- Often indicative of a material's **properties**
- Can be important **appearance cue**, especially if shape is similar across objects
- Distinguish between shape, boundaries, and texture

Technically:

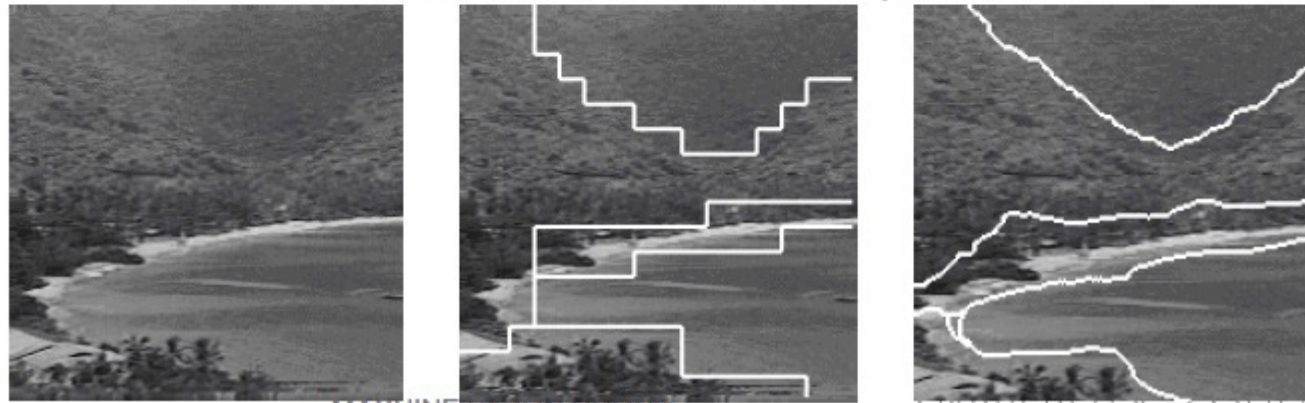
- Representation-wise, we want a feature one step above “building blocks” of filters, edges.

Why analyze texture?

Natural scene #1: 384x384 pixels



Natural scene #2: 192x192 pixels

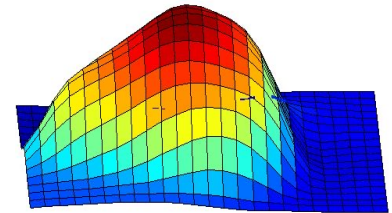
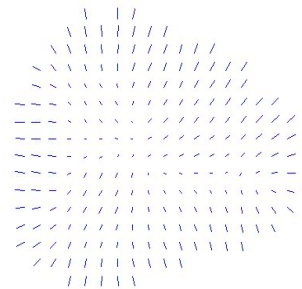
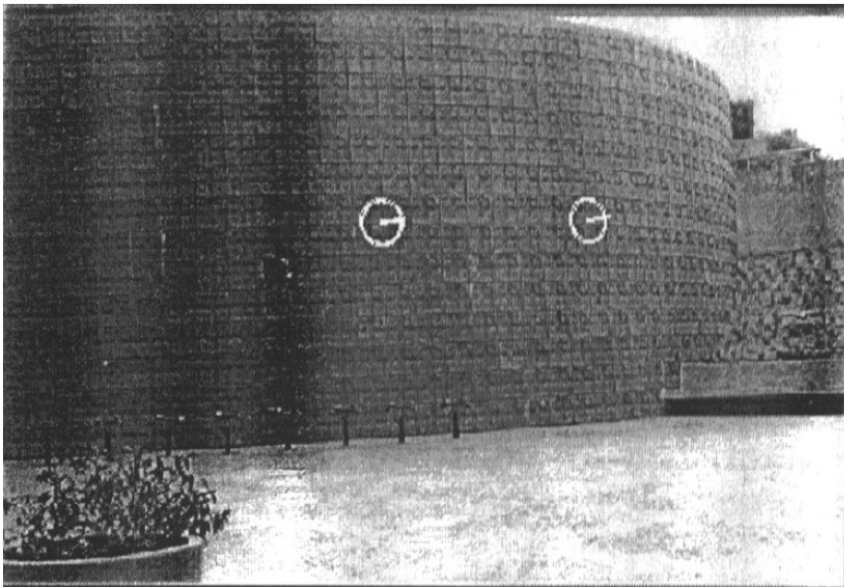


Texture-related tasks

- **Shape from texture**
 - Estimate surface orientation or shape from image texture
- **Segmentation/classification** from texture cues
 - Analyze, represent texture
 - Group image regions with consistent texture
- **Synthesis**
 - Generate new texture patches/images given some examples

Shape from texture

- Use deformation of texture from point to point to estimate surface shape

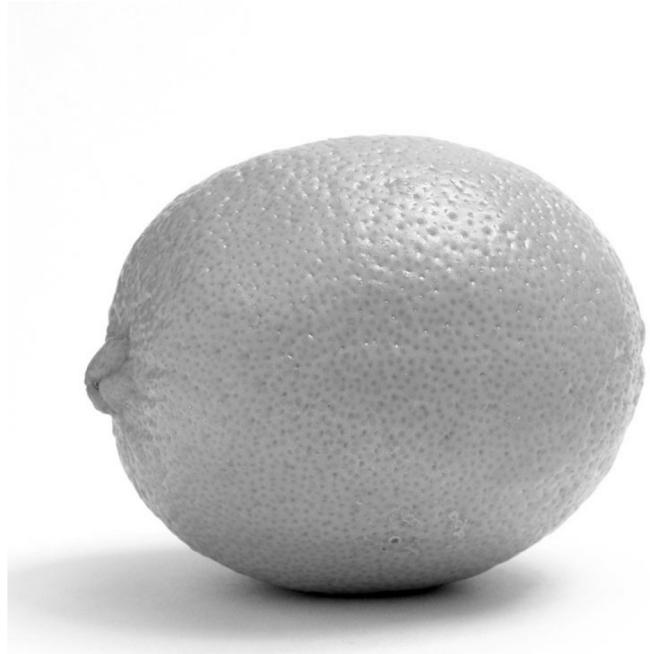




Texture-related tasks

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Texture-related tasks



Texture-related tasks



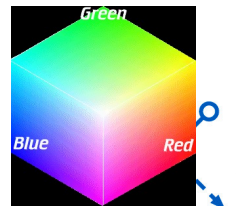
Color vs. texture

- Recall: These looked very similar in terms of their color distributions (when our features were R-G-B)
- But how would their texture distributions compare?

query



query



Psychophysics of texture

- Some textures distinguishable with *pre-attentive* perception– without scrutiny, eye movements [Julesz 1975]

Watch the next screen very carefully.

Are the **left** and **right** sides:

Same or different?

Pre-attentive processing is the subconscious accumulation of information from the environment. All available information is pre-attentively processed.

Ready? First Example

Left

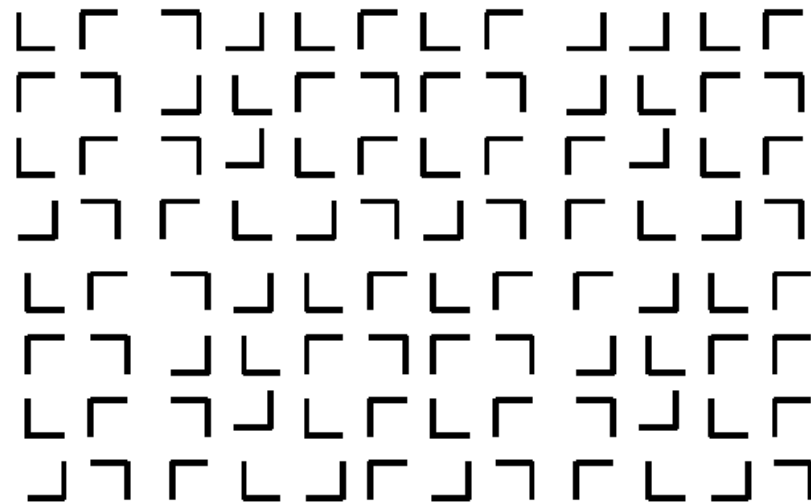
Right

L	┐	┌	J	L	┐	┐	┌	T	┐	┐	┌
┐	┌	J	L	┐	┌	┌	T	┌	┌	┌	T
L	┐	┌	J	L	┐	┌	┌	T	┌	┌	T
J	┌	┐	L	J	┌	┐	T	T	┌	┐	T
L	┐	┌	J	L	┐	┐	T	┌	┐	┌	┌
┐	┌	J	L	┐	┌	┌	T	┐	┌	┌	T
L	┐	┌	J	L	┐	┐	┌	┌	┐	┐	┌
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Ready? Second Example

Left

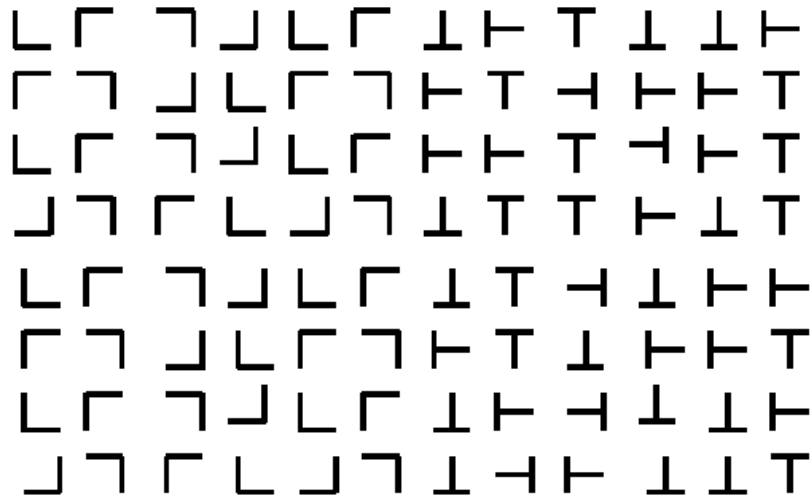
Right



What you have seen

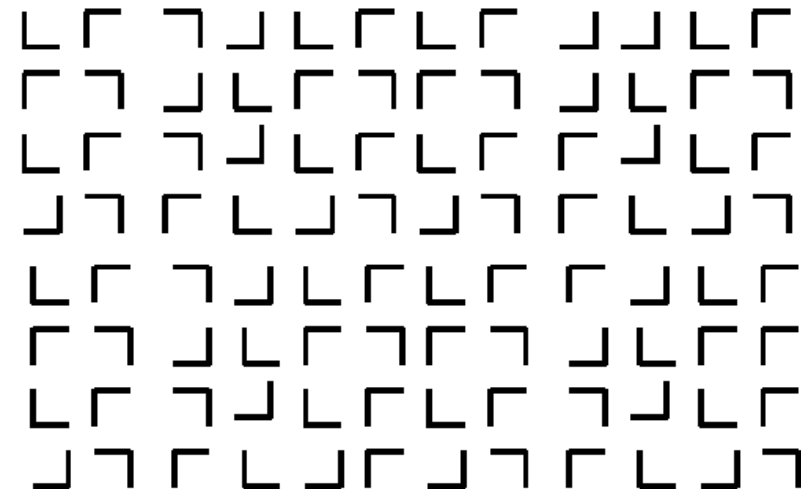
Left

Right



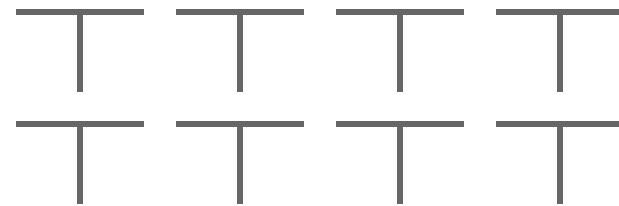
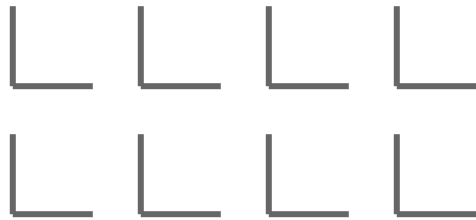
Left

Right



Texture Analysis

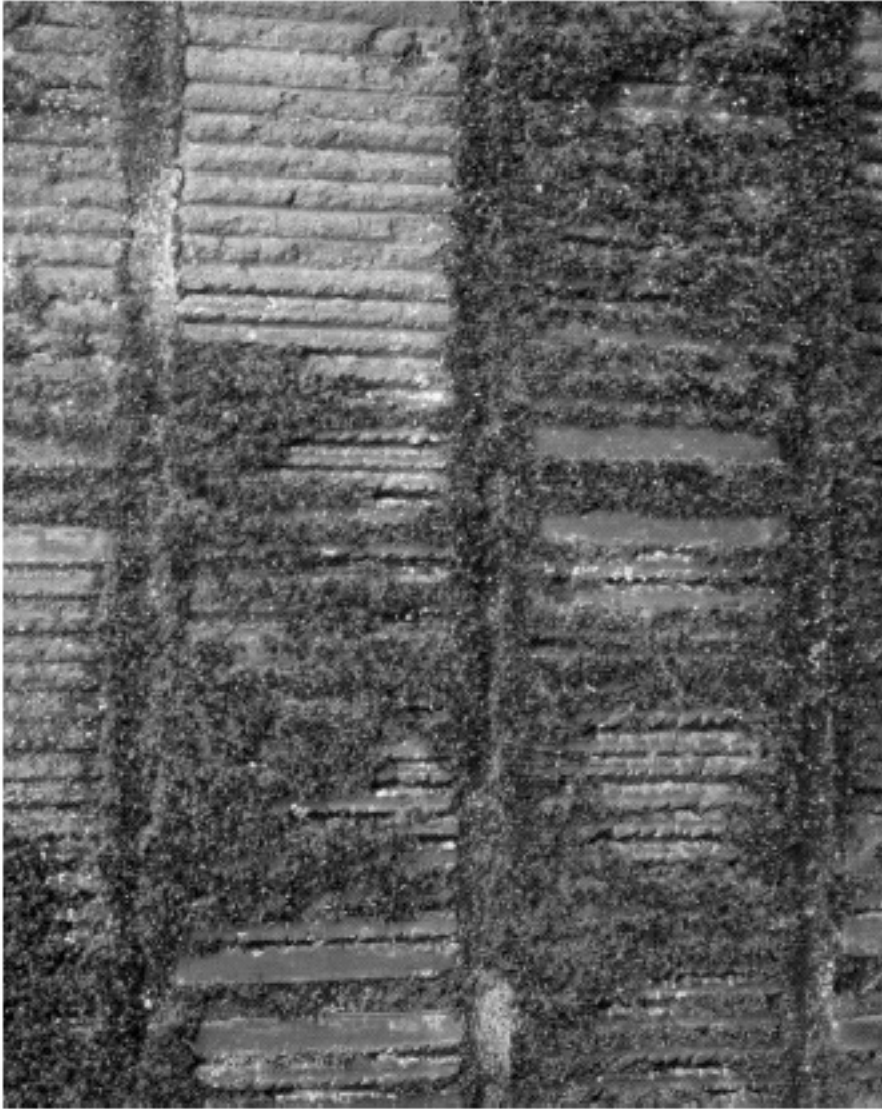
- Analyze the texture in terms of statistical relationships between fundamental texture elements
 - Called “textons”.
- It generally required a human to look at the texture in order to decide what those fundamental units were.



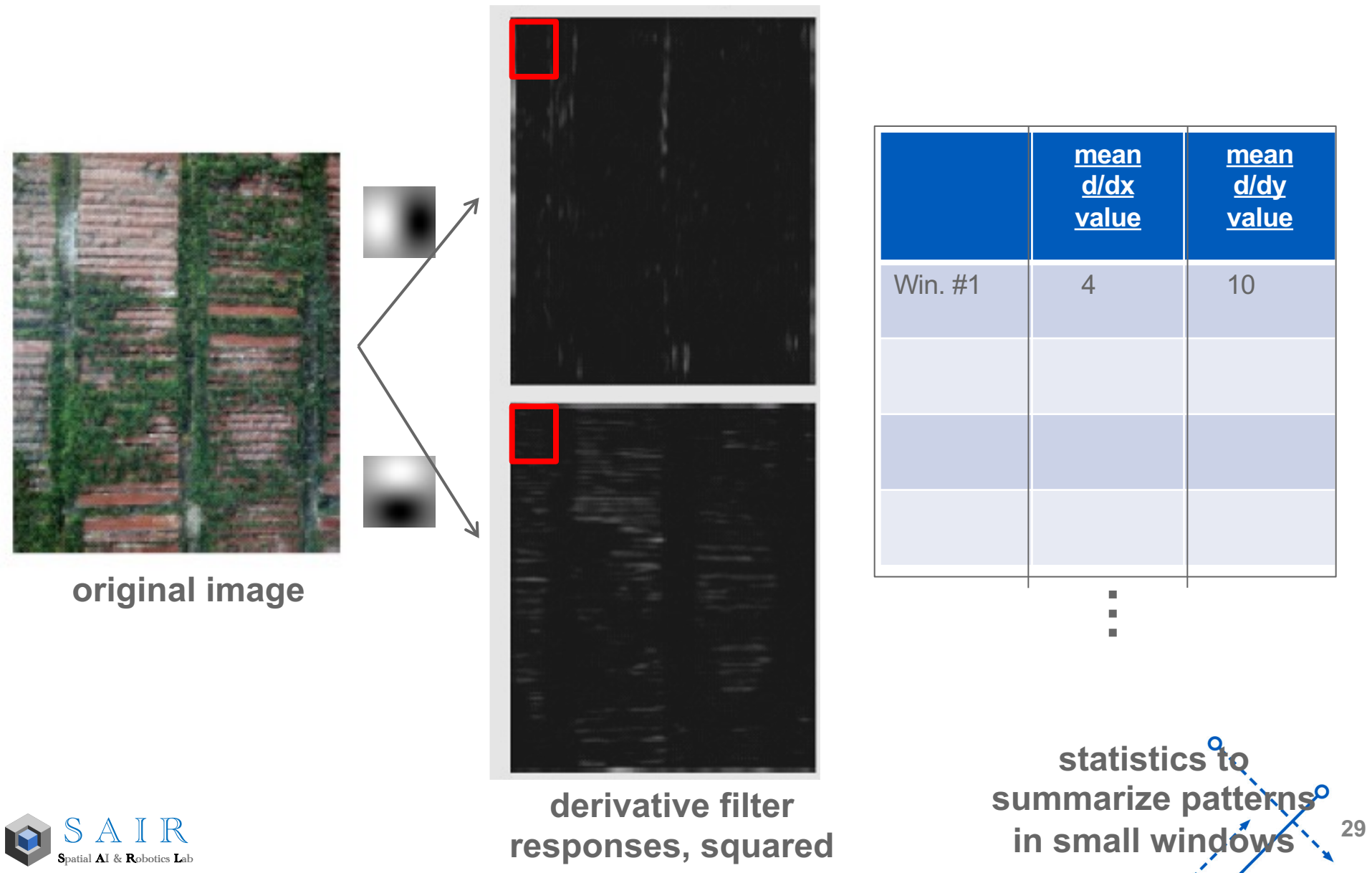
Texture representation

- Textures are made up of repeated local patterns:
 - How to find the patterns
 - Use filters that look like patterns
 - Spots, bars, raw patches...
 - Consider magnitude of response
 - Describe their statistics within each local window
 - Mean, standard deviation, etc.
 - Histogram of “prototypical” feature occurrences

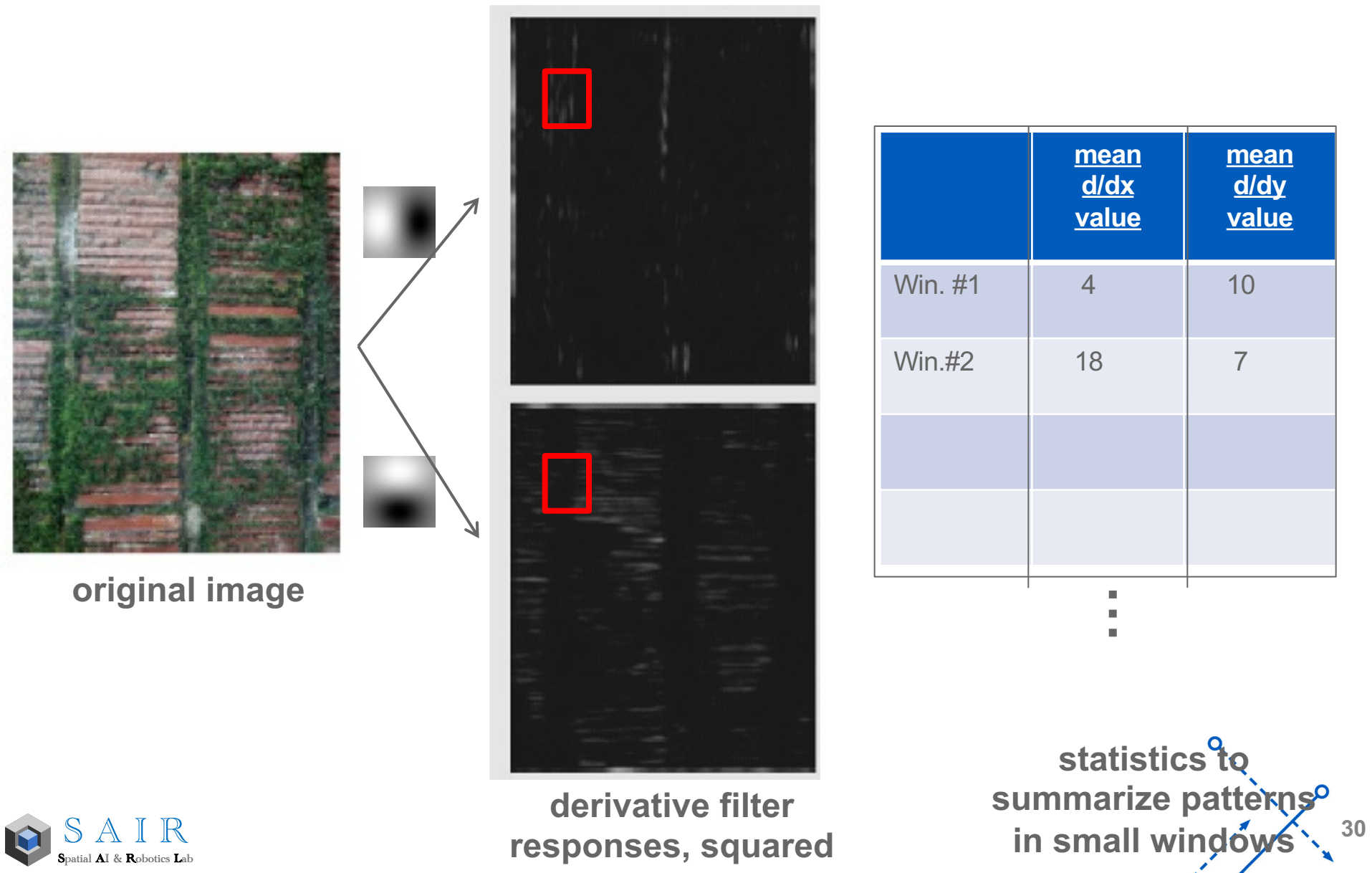
Texture representation



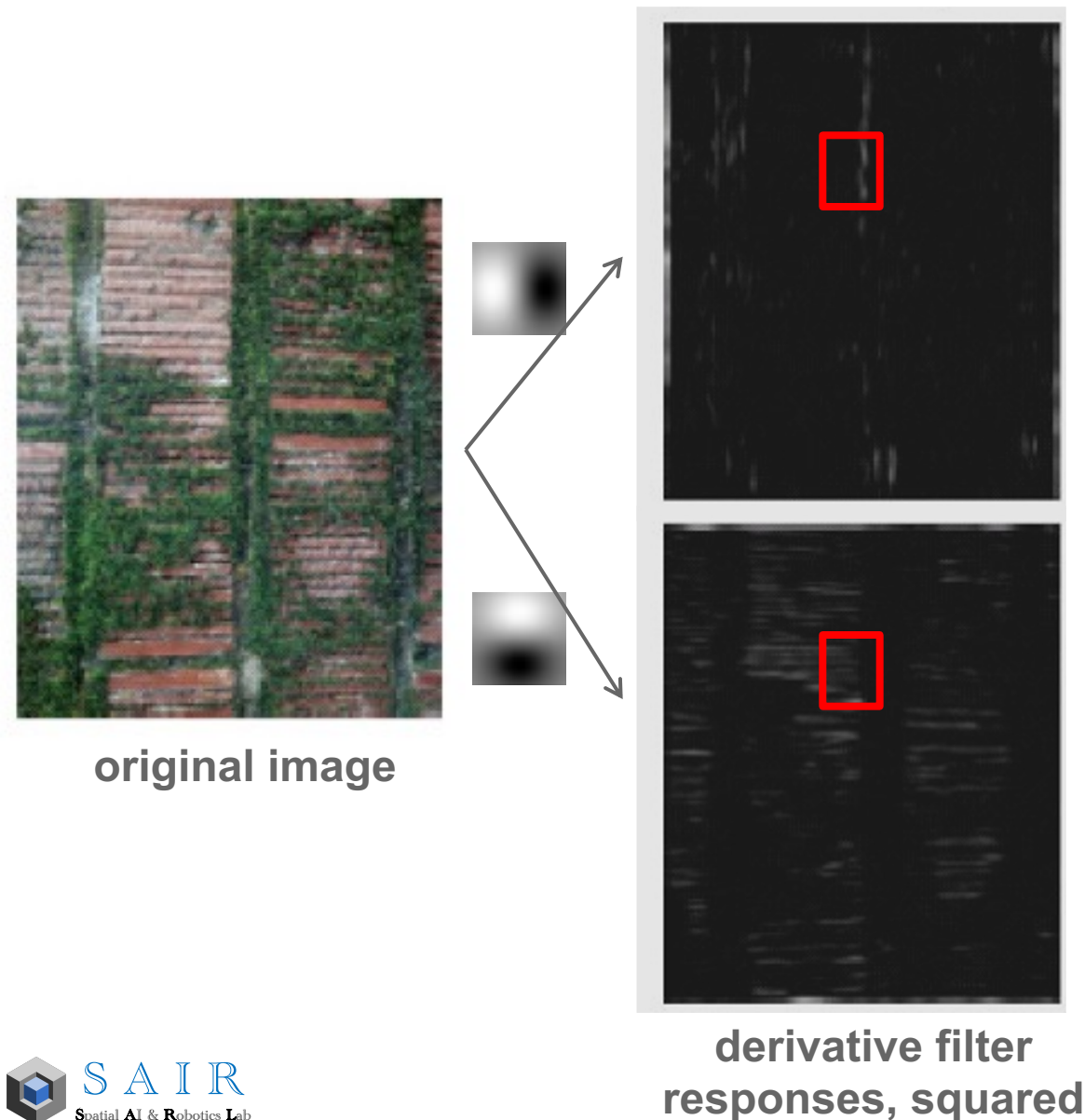
Texture representation



Texture representation



Texture representation



	<u>mean d/dx value</u>	<u>mean d/dy value</u>
Win. #1	4	10
Win.#2	18	7
⋮		
Win.#9	20	20

⋮

statistics to
summarize patterns
in small windows



-
-
-

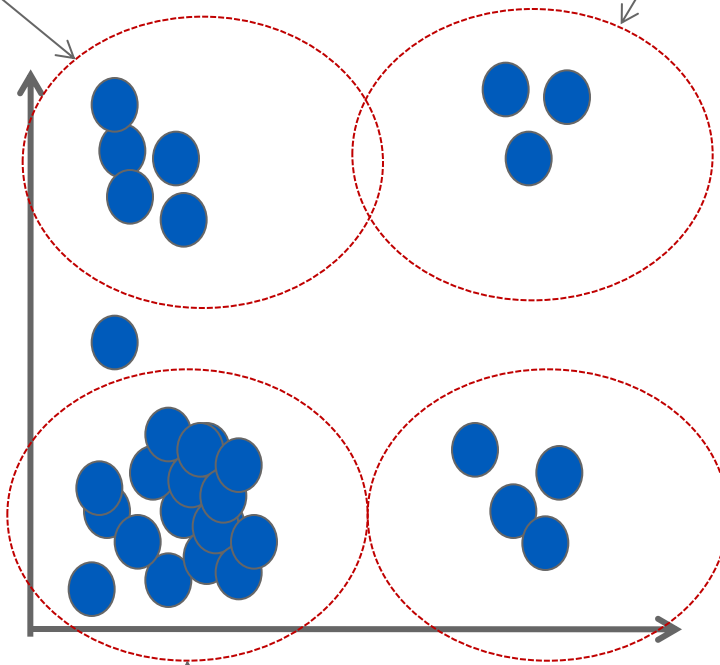
s to
patterns
ndows

Texture representation: example

Windows with
primarily horizontal
edges

Both

Dimension 2 (mean d/dy value)



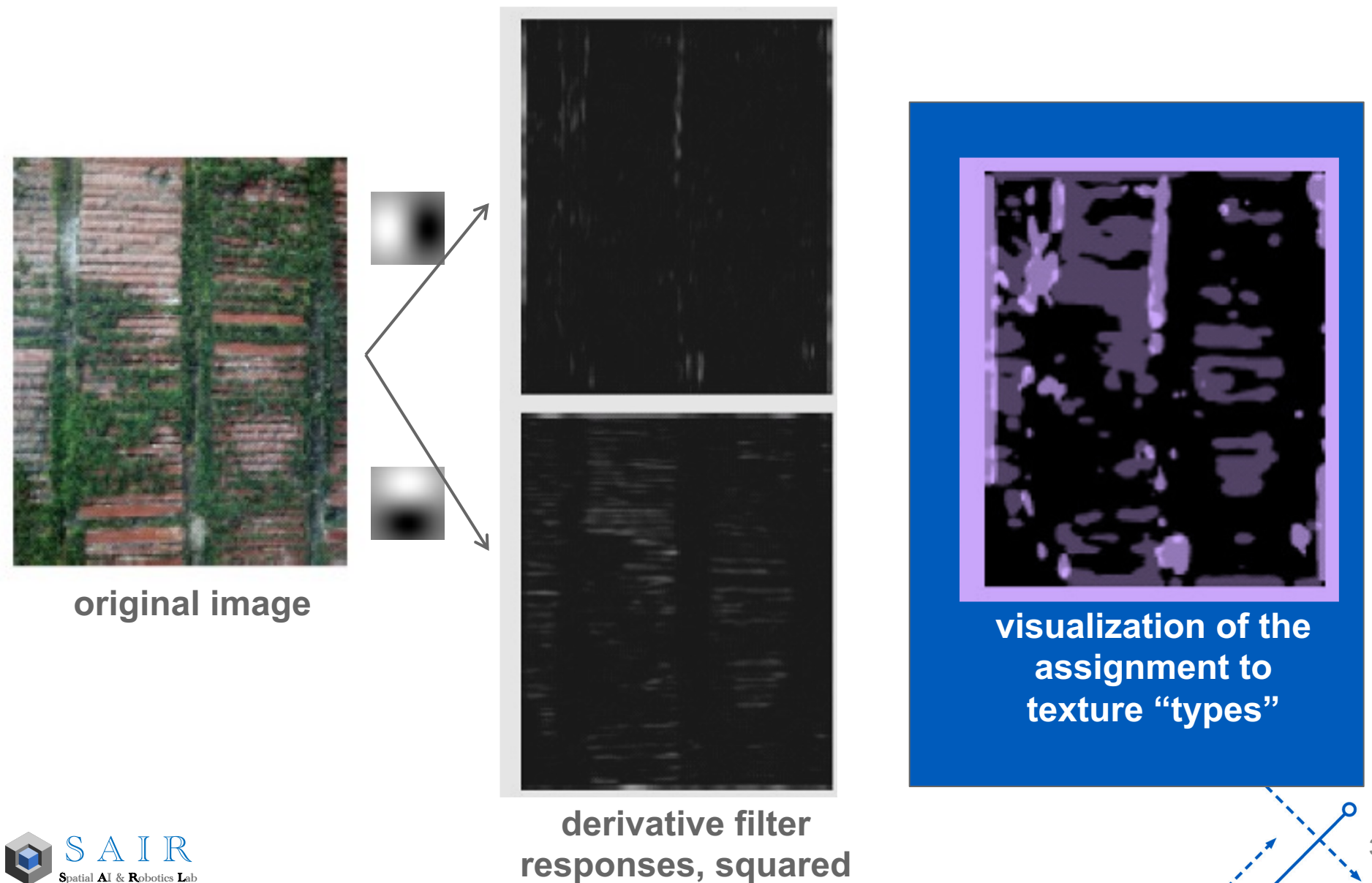
Windows with
small gradient in
both directions

Windows with
primarily vertical
edges

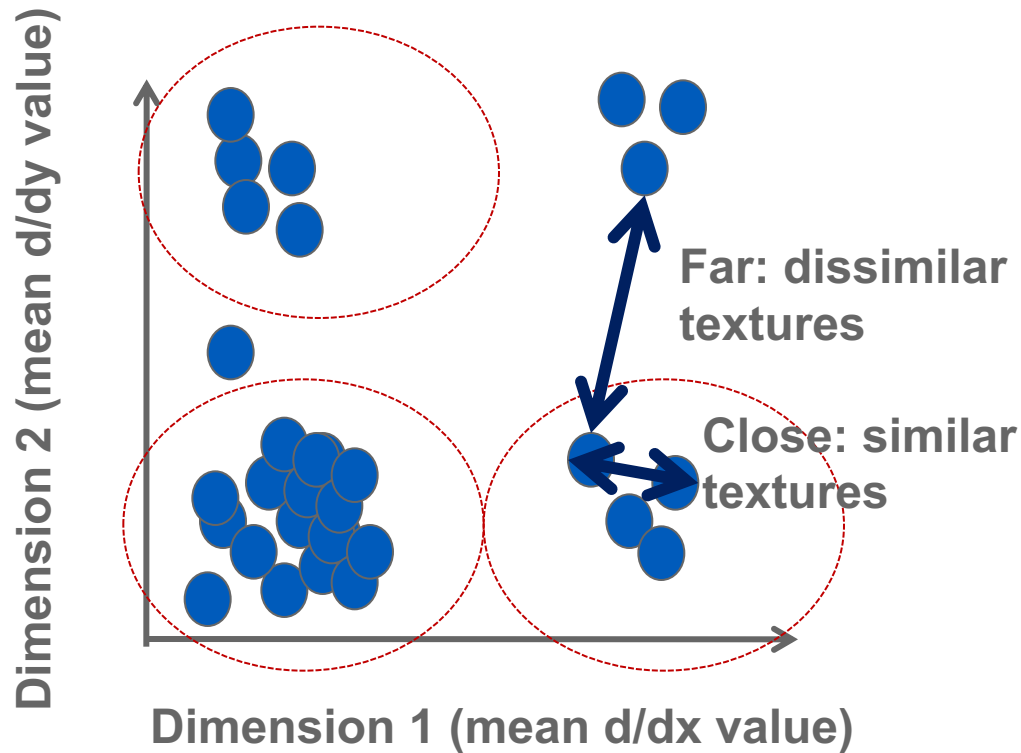
	<u>mean d/dx value</u>	<u>mean d/dy value</u>
Win. #1	4	10
Win.#2	18	7
⋮		
Win.#9	20	20
	⋮	

statistics to
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Texture representation: example



Texture representation: example

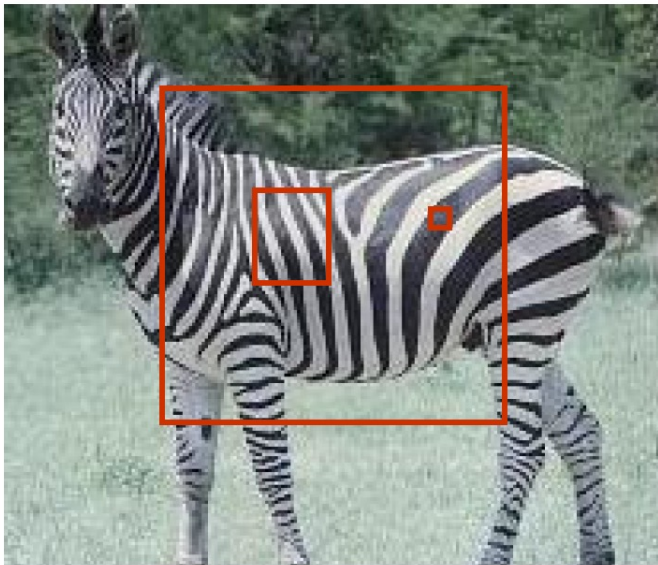


	<u>mean d/dx value</u>	<u>mean d/dy value</u>
Win. #1	4	10
Win.#2	18	7
⋮		
Win.#9	20	20
	⋮	

statistics to
summarize patterns
in small windows

Texture representation: window scale

- We're assuming we know the relevant window size for which we collect these statistics.

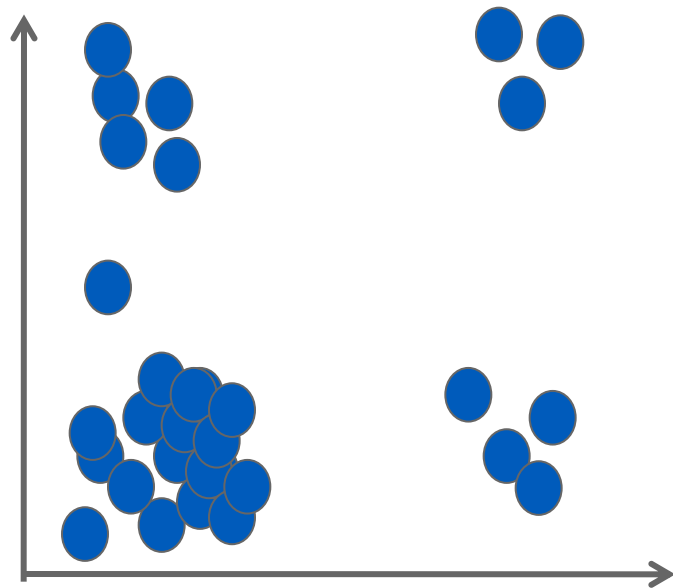


Possible to perform scale selection by looking for window scale where texture description not changing.

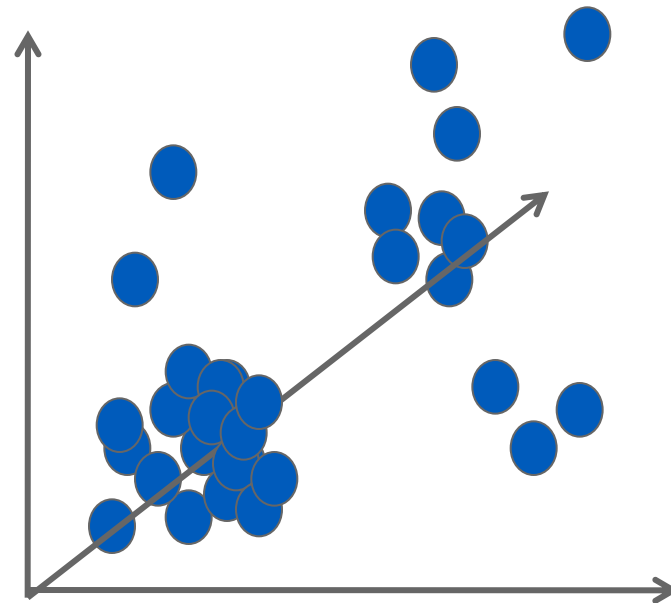
Filter banks

- Our previous example used two filters, and resulted in a 2-dimensional feature vector to describe texture in a window.
 - x and y derivatives revealed something about local structure.
- We can generalize to apply a collection of multiple (d) filters: a “filter bank”
 - Recall multi-channel convolution.
- Then our feature vectors will be d -dimensional.
 - still can think of nearness, farness in feature space

d -dimensional features



2d

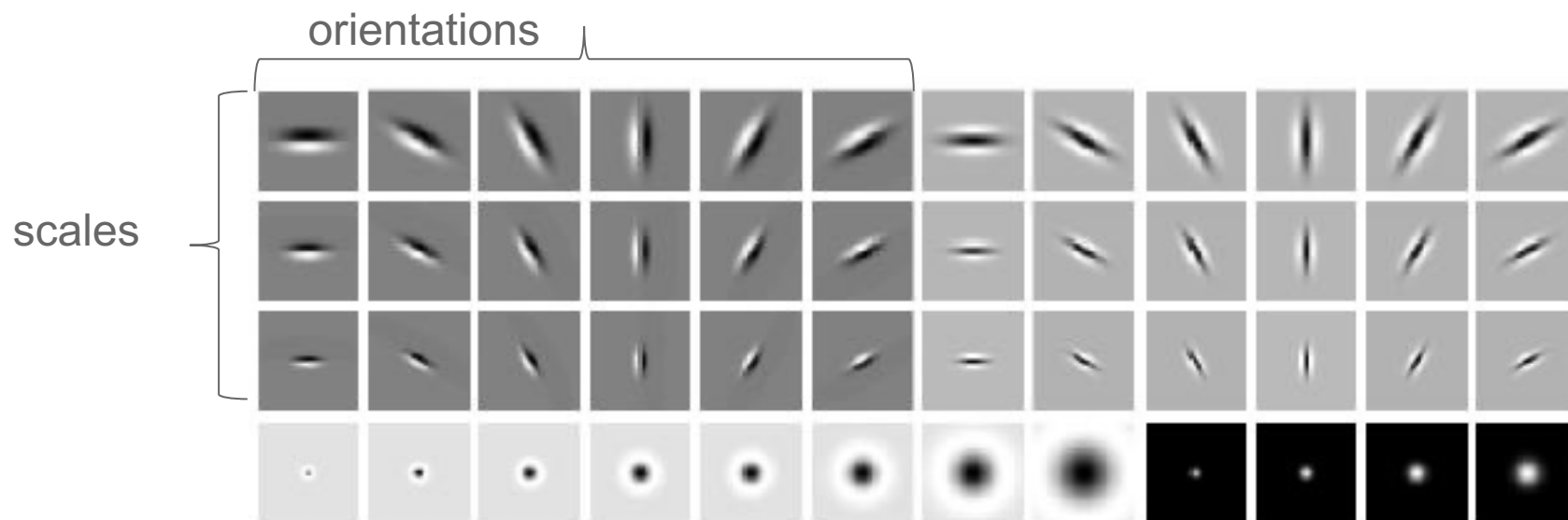


3d

...

Filter banks

- What filters to put in the bank?
 - Typically we want a combination of scales and orientations, different types of patterns.



Pixel Neighborhood-based Feature

- The most important for texture analysis is to describe the spatial behavior of intensity values in any neighborhood.
- Different methodologies have been proposed.
- **Local binary pattern (LBP)** is one of the most-widely used approach – mainly for face recognition.
- LBP is used for texture analysis too.

Local Binary Pattern (LBP)

For each PIXEL of an image, a BINARY CODE is produced

$$LBP_{p,r}(N_c) = \sum_{p=0}^{P-1} g(N_p - N_c) 2^p$$

where

neighborhood pixels (N_p) in each block $\rightarrow$ is thresholded by its center pixel (N_c)

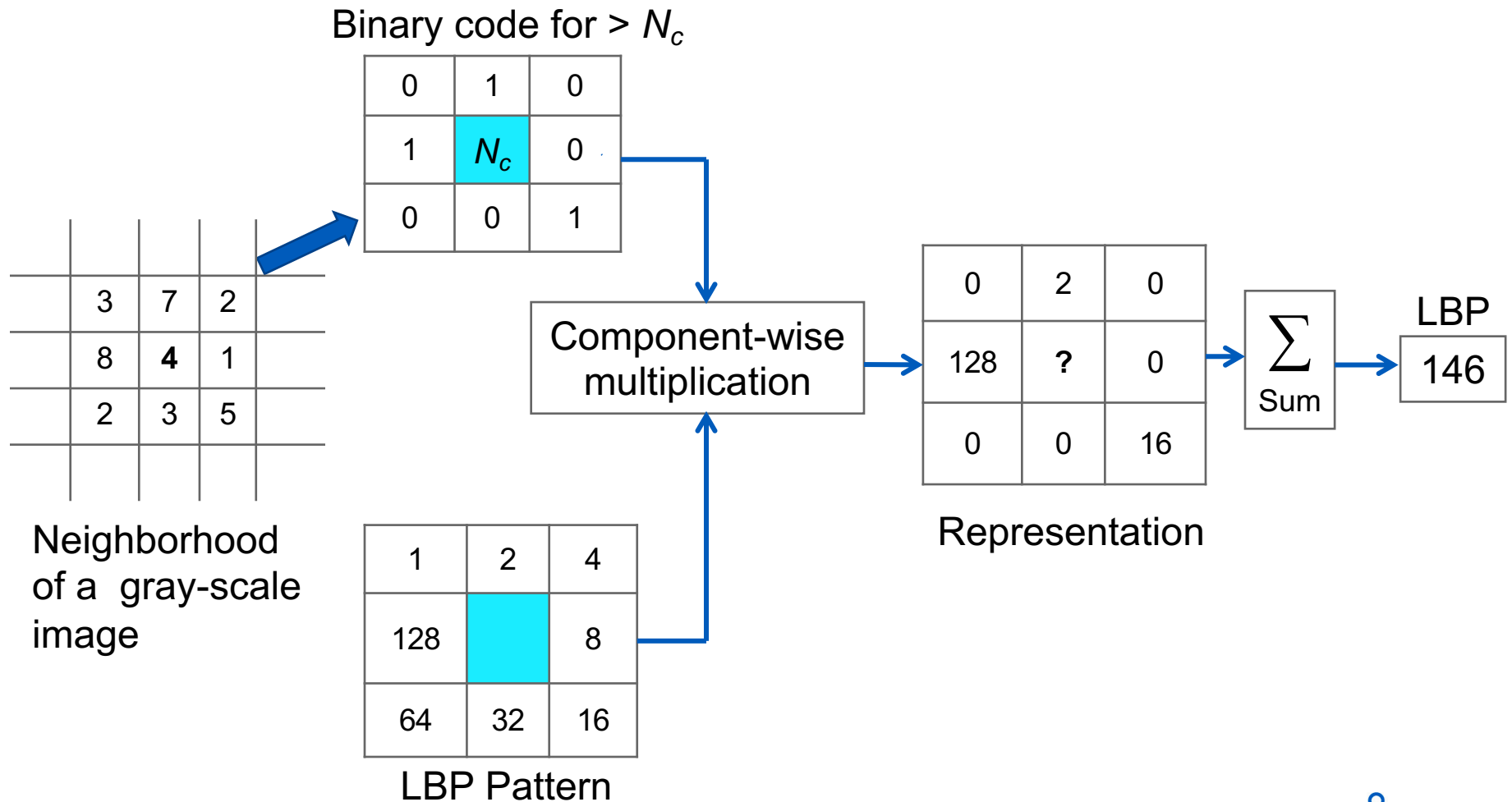
$p \rightarrow$ sampling points (e.g., $p = 0, 1, \dots, 7$ for a 3x3 cell, where $P = 8$)

$r \rightarrow$ radius (for 3x3 cell, it is 1).

Binary threshold function $g(x)$ is,

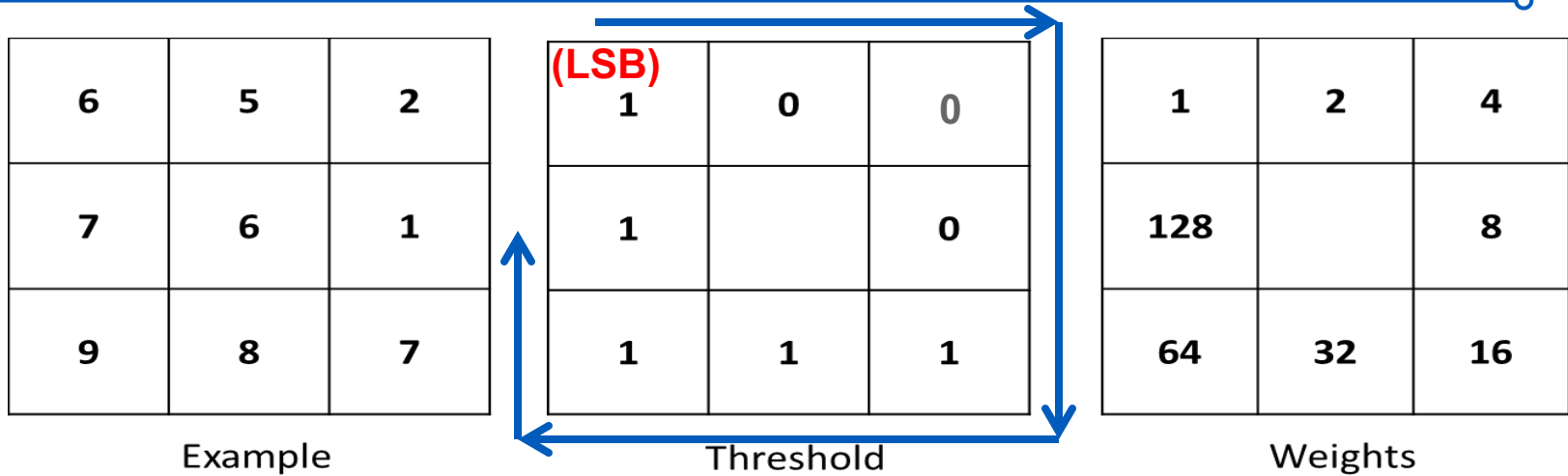
$$g(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

Computation of Local Binary Pattern



Example of how the *LBP operator* works

Computation of Local Binary Pattern



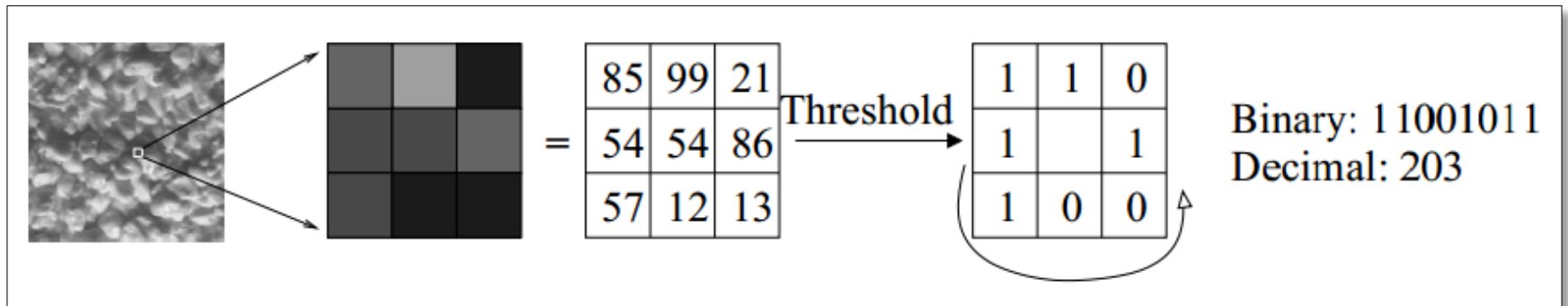
Binary Pattern:	1 (MSB)	1	1	1	0	0	0	1 (LSB)
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Code/Weight (2^p):	1×2^7	1×2^6	1×2^5	1×2^4	0×2^3	0×2^2	0×2^1	1×2^0
	= 128	= 64	= 32	= 16	= 0	= 0	= 0	= 1

LBP:	$128 + 64 + 32 + 16 + 0 + 0 + 0 + 1 = 241$
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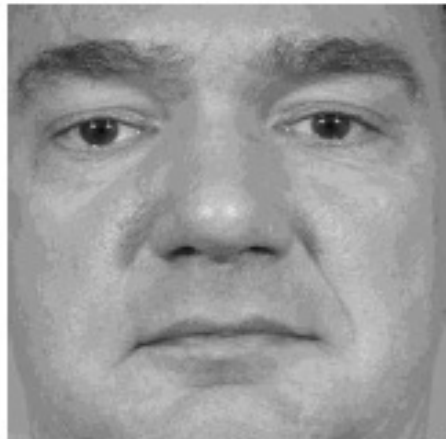
LBP

- One of the best performing texture descriptors
- A label is assigned to every pixel
- Use center pixel to threshold the 3x3 neighborhood
 - Result in binary number
- Histogram of the labels is used as a texture descriptor



The Basic LBP operator

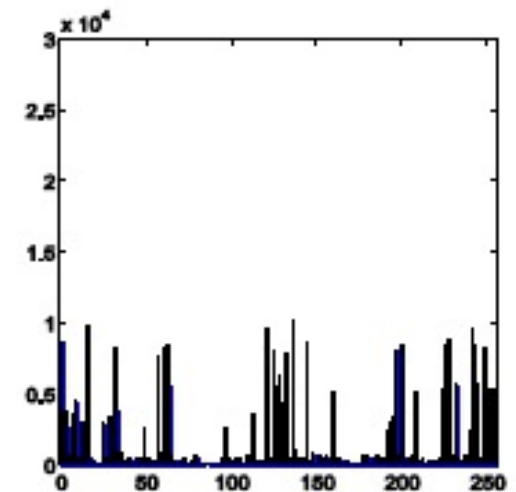
LBP histogram



Input image



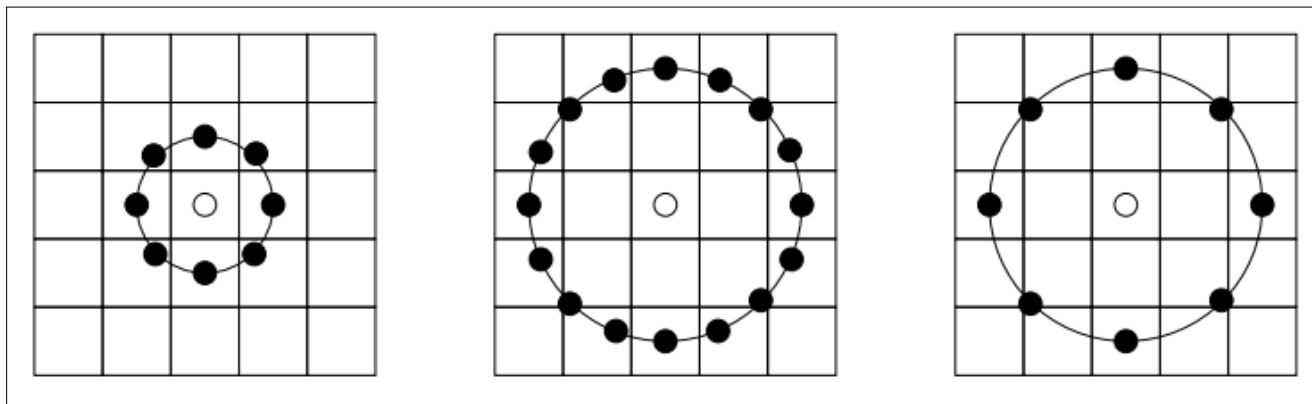
LBP image



LBP histogram

LBP

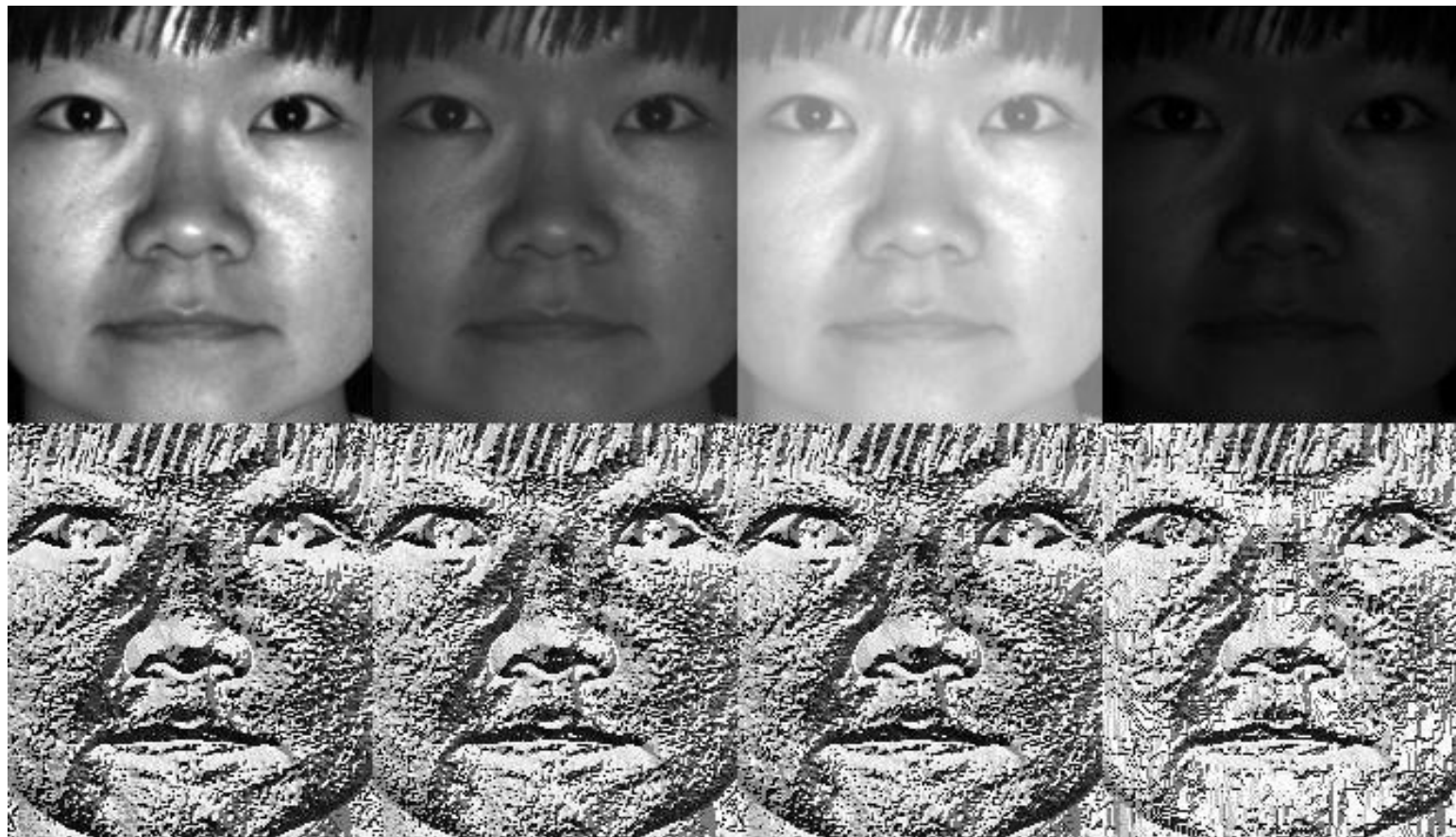
- LBP is extended to use different sizes of neighborhoods.
- Local neighborhoods is defined as a set of **sampling points**.
 - points evenly spaced on a circle centered at the labeled pixel.
- **(P, R)** , **P = number of sampling points**, **R = radius**
- **Bilinear interpolation** is used
 - If sampling point is not in the center of the pixel.



The circular (8,1), (16, 2) and (8, 2) neighborhoods.

LBP Example

- Notice how LBP feature are illumination invariant



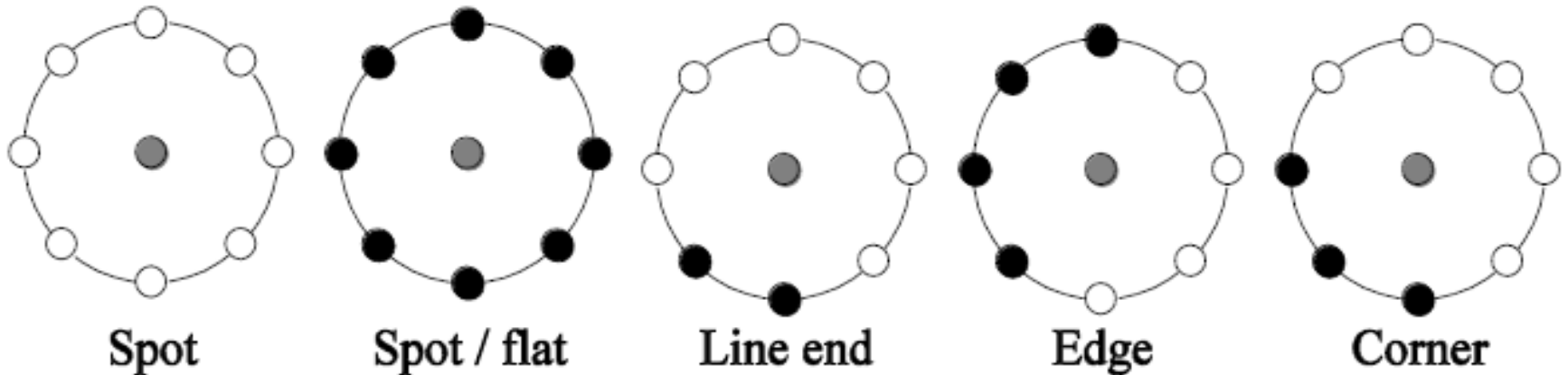
Uniform LBP

- Uniform patterns to further improve LBP.
 - Has at most 2 bitwise transitions in binary pattern.
 - Histogram
 - assigns separate bins for every uniform pattern.
 - assigns a single bin for all non-uniform pattern.

Uniform LBP

Texture primitives (“micro-textons”) detected by the uniform patterns of LBP

1 = black
0 = white



Uniform LBP

- Uniform patterns examples
 - 00000000 (0 transitions)
 - 01110000 (2 transitions)
 - 11001111 (2 transitions)
- Non-uniform patterns examples
 - 11001001 (4 transitions)
 - 01010011 (5 transitions)
- 59 bins histogram
 - 58 uniform bins →
 - 1 non-uniform bin

Length of feature vector

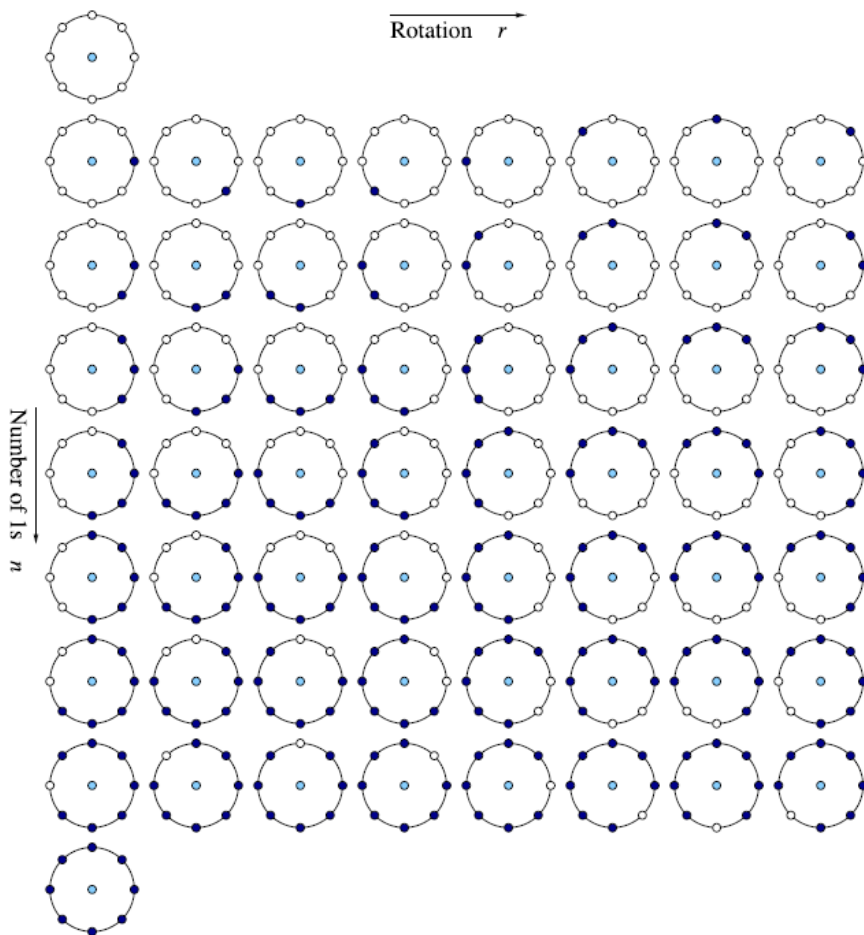
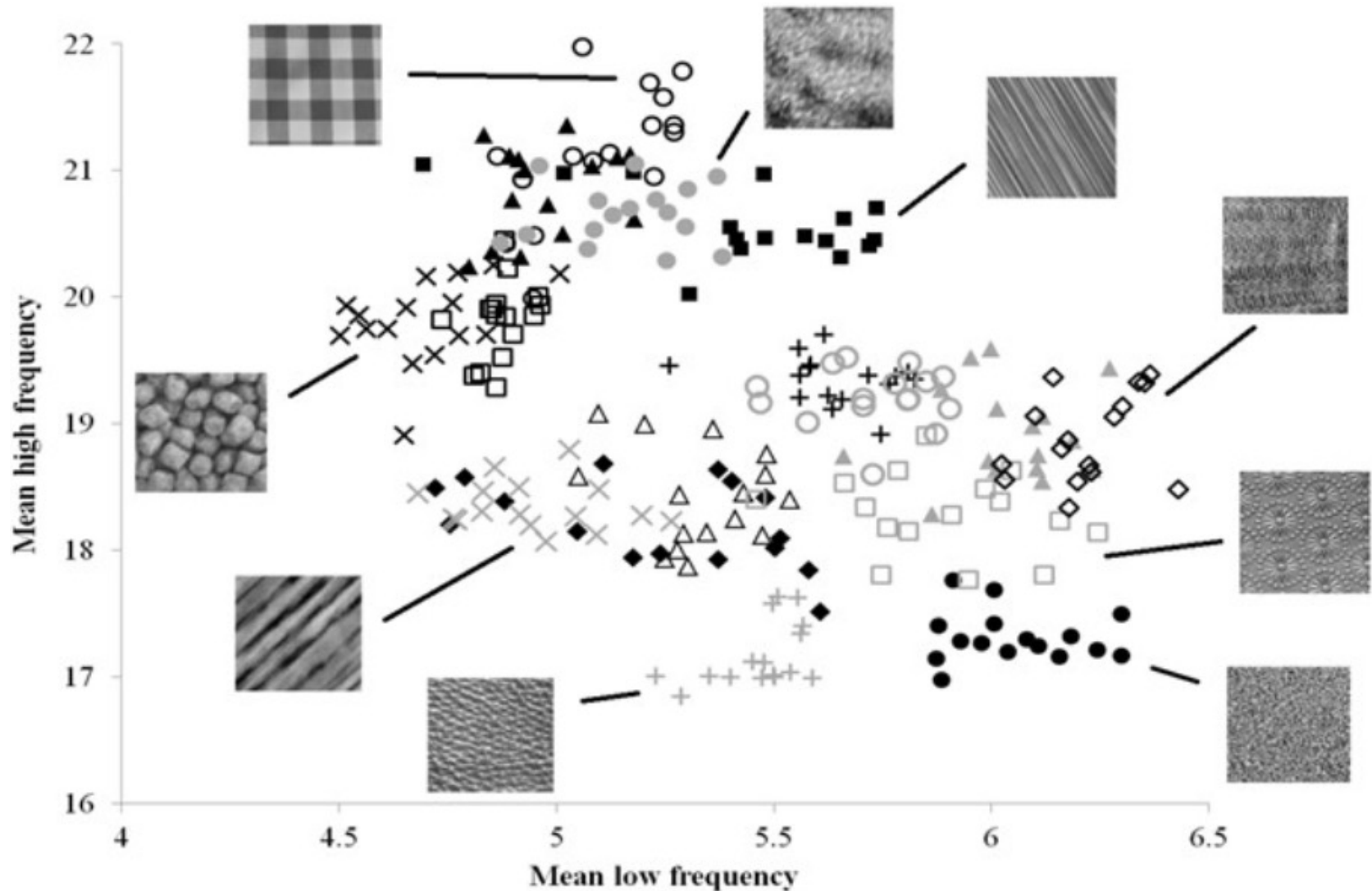


Fig. 2.4 The 58 different uniform patterns in $(8, R)$ neighborhood

Other Measures of Texture (frequency)



Other Measures of Texture

Texture measure	Description	Equation*
Mean	Mean is the average grey level for each sample	$\sum_{i,j=0}^{N-1} i(P_{i,j})$
Variance	Variance is a measure of heterogeneity; it increases when the grey level values differ from their mean. In general, coarse-textured features are associated with higher variances	$\sum_{i,j=0}^{N-1} P_{i,j}(i-\mu_i)^2$
Homogeneity	This parameter measures image homogeneity as it assumes larger values for smaller grey tone differences in pair elements	$\sum_{i,j=0}^{N-1} \frac{P_{i,j}}{1+(i-j)^2}$
Contrast	Contrast is a measure of spatial frequency, the difference between the highest and the lowest values of a contiguous set of pixels. A high contrast implies high coarse texture	$\sum_{i,j=0}^{N-1} P_{i,j}(i-j)^2$
Dissimilarity	Dissimilarity, akin to contrast, describes the heterogeneity of the grey levels. Higher values of dissimilarity in the GLC matrix indicate coarser textures	$\sum_{i,j=0}^{N-1} P_{i,j} i-j $
Entropy	Entropy measures the disorder of an image. When the image is not texturally uniform, many GLC matrix elements have very small values, which imply that entropy is very large	$\sum_{i,j=0}^{N-1} P_{i,j}(-\ln p_{i,j})^2$
Angular second moment	This parameter measures textural uniformity. Thus, high angular second moment values occur when the grey level distribution over the window has either a constant or a periodic form	$\sum_{i,j=0}^{N-1} P_{i,j}^2$
Correlation	Correlation is a measure of grey-tone linear dependencies in the image. High correlation values imply a linear relationship between the grey levels of pixel pairs	$\sum_{i,j=0}^{N-1} P_{i,j} \left[\frac{(i-\mu_i)(j-\mu_j)}{\sqrt{(SD_i^2)(SD_j^2)}} \right]$

